

```
import numpy as np
import pandas as pd
```

▼ Creating DataFrame

```
# using lists
student_data = [
    [100,80,10],
    [90,70,7],
    [120,100,14],
    [80,50,2]
]

pd.DataFrame(student_data,columns=['iq','marks','package'])
```



	iq	marks	package
0	100	80	10
1	90	70	7
2	120	100	14
3	80	50	2

```
# using dicts

student_dict = {
    'name':['nitish','ankit','rupesh','rishabh','amit','ankita'],
    'iq':[100,90,120,80,0,0],
    'marks':[80,70,100,50,0,0],
    'package':[10,7,14,2,0,0]
}

students = pd.DataFrame(student_dict)
students.set_index('name',inplace=True)
students
```



	iq	marks	package
name			
nitish	100	80	10
ankit	90	70	7
rupesh	120	100	14
rishabh	80	50	2
amit	0	0	0
ankita	0	0	0

```
# using read_csv
movies = pd.read_csv('movies.csv')
movies
```

		title_x	imdb_id	poster_path	wiki_link	title_y	original_title
0		Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike
1		Battalion 609	tt9472208	NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609
2		The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister
3		Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India
4		Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows
...		...	...	...	...	...	...
1624		Tera Mera Saath Rahen	tt0301250	https://upload.wikimedia.org/wikipedia/en/2/2b...	https://en.wikipedia.org/wiki/Tera_Mera_Saath_...	Tera Mera Saath Rahen	Tera Mera Saath Rahen
1625		Yeh Zindagi Ka Safar	tt0298607	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Zindagi_Ka_S...	Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar
1626		Sabse Bada Sukh	tt0069204	NaN	https://en.wikipedia.org/wiki/Sabse_Bada_Sukh	Sabse Bada Sukh	Sabse Bada Sukh
1627		Daaka	tt10833860	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Daaka	Daaka	Daaka

```
ipl = pd.read_csv('ipl-matches.csv')
ipl
```



	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	WinningTe
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat	N	Guja Tita
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field	N	Rajasth Roy.
2	1312198	Kolkata	2022-05-25	2022	Eliminator	Royal Challengers Bangalore	Lucknow Super Giants	Eden Gardens, Kolkata	Lucknow Super Giants	field	N	Ro Challenge Bangal
3	1312197	Kolkata	2022-05-24	2022	Qualifier 1	Rajasthan Royals	Gujarat Titans	Eden Gardens, Kolkata	Gujarat Titans	field	N	Guja Tita
4	1304116	Mumbai	2022-05-22	2022	70	Sunrisers Hyderabad	Punjab Kings	Wankhede Stadium, Mumbai	Sunrisers Hyderabad	bat	N	Punjab Kin
...	...	...	...	...	...	...	...	...	...	...	...	...
945	335986	Kolkata	2008-04-20	2007/08	4	Kolkata Knight Riders	Deccan Chargers	Eden Gardens	Deccan Chargers	bat	N	Kolka Knight Ride
946	335985	Mumbai	2008-04-20	2007/08	5	Mumbai Indians	Royal Challengers Bangalore	Wankhede Stadium	Mumbai Indians	bat	N	Ro Challenge Bangal
947	335984	Delhi	2008-04-19	2007/08	3	Delhi Daredevils	Rajasthan Royals	Feroz Shah Kotla	Rajasthan Royals	bat	N	De Daredev
948	335983	Chandigarh	2008-04-19	2007/08	2	Kings XI Punjab	Chennai Super Kings	Punjab Cricket Association Stadium, Mohali	Chennai Super Kings	bat	N	Cheni Super Kin
949	335982	Bangalore	2008-04-18	2007/08	1	Royal Challengers Bangalore	Kolkata Knight Riders	M Chinnaswamy Stadium	Royal Challengers Bangalore	field	N	Kolka Knight Ride

▼ DataFrame Attributes and Methods

```
# shape
movies.shape
ipl.shape

 (950, 20)

# dtypes
movies.dtypes
ipl.dtypes

 ID          int64
City         object
Date         object
Season       object
MatchNumber  object
Team1        object
Team2        object
Venue        object
TossWinner   object
TossDecision object
```

```

SuperOver      object
WinningTeam    object
WonBy          object
Margin         float64
method         object
Player_of_Match object
Team1Players   object
Team2Players   object
Umpire1        object
Umpire2        object
dtype: object

```

```

# index
movies.index
ipl.index

```

```
↩ RangeIndex(start=0, stop=950, step=1)
```

```

# columns
movies.columns
ipl.columns
student.columns

```

```
↩ Index(['iq', 'marks', 'package'], dtype='object')
```

```

# values
student.values
ipl.values

```

```

↩ array([[1312200, 'Ahmedabad', '2022-05-29', ...,
        "['WP Saha', 'Shubman Gill', 'MS Wade', 'HH Pandya', 'DA Miller', 'R Tewatia', 'Rashid Khan', 'R Sai Kishore', 'LH Ferguson',
        'Yash Dayal', 'Mohammed Shami']]",
        'CB Gaffaney', 'Nitin Menon'],
        [1312199, 'Ahmedabad', '2022-05-27', ...,
        "['YBK Jaiswal', 'JC Buttler', 'SV Samson', 'D Padikkal', 'SO Hetmyer', 'R Parag', 'R Ashwin', 'TA Boult', 'YS Chahal', 'M
        Prasad Krishna', 'OC McCoy']]",
        'CB Gaffaney', 'Nitin Menon'],
        [1312198, 'Kolkata', '2022-05-25', ...,
        "['Q de Kock', 'KL Rahul', 'M Vohra', 'DJ Hooda', 'MP Stoinis', 'E Lewis', 'KH Pandya', 'PVD Chameera', 'Mohsin Khan', 'Avesh
        Khan', 'Ravi Bishnoi']]",
        'J Madanagopal', 'MA Gough'],
        ...,
        [335984, 'Delhi', '2008-04-19', ...,
        "['T Kohli', 'YK Pathan', 'SR Watson', 'M Kaif', 'DS Lehmann', 'RA Jadeja', 'M Rawat', 'D Salunkhe', 'SK Warne', 'SK Trivedi',
        'MM Patel']]",
        'Aleem Dar', 'GA Pratapkumar'],
        [335983, 'Chandigarh', '2008-04-19', ...,
        "['PA Patel', 'ML Hayden', 'MEK Hussey', 'MS Dhoni', 'SK Raina', 'JDP Oram', 'S Badrinath', 'Joginder Sharma', 'P Amarnath',
        'MS Gony', 'M Muralitharan']]",
        'MR Benson', 'SL Shastri'],
        [335982, 'Bangalore', '2008-04-18', ...,
        "['SC Ganguly', 'BB McCullum', 'RT Ponting', 'DJ Hussey', 'Mohammad Hafeez', 'LR Shukla', 'WP Saha', 'AB Agarkar', 'AB Dinda',
        'M Kartik', 'I Sharma']]",
        'Asad Rauf', 'RE Koertzen']], dtype=object)

```

```

# head and tail
movies.head(2)

```

```

↩

```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title	is_adul
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thumb/...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike	1
1	Battalion 609	tt9472208	NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609	1

```
ipl.tail(2)
```



	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	WinningTeam	I
948	335983	Chandigarh	2008-04-19	2007/08	2	Kings XI Punjab	Chennai Super Kings	Punjab Cricket Association Stadium, Mohali	Chennai Super Kings	bat	N	Chennai Super Kings	
949	335982	Bangalore	2008-04-18	2007/08	1	Royal Challengers Bangalore	Kolkata Knight Riders	M Chinnaswamy Stadium	Royal Challengers Bangalore	field	N	Kolkata Knight Riders	

```
# sample
ipl.sample(5)
```



	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	WinningTeam	
336	1082628	Mumbai	2017-05-01	2017	38	Mumbai Indians	Royal Challengers Bangalore	Wankhede Stadium	Royal Challengers Bangalore	bat	N	Mumbai Indians	
98	1254107	Sharjah	2021-09-25	2021	37	Punjab Kings	Sunrisers Hyderabad	Sharjah Cricket Stadium	Sunrisers Hyderabad	field	N	Punjab Kings	
890	392182	Cape Town	2009-04-18	2009	2	Royal Challengers Bangalore	Rajasthan Royals	Newlands	Royal Challengers Bangalore	bat	N	Royal Challengers Bangalore	
157	1216533	Abu Dhabi	2020-10-19	2020/21	37	Chennai Super Kings	Rajasthan Royals	Sheikh Zayed Stadium	Chennai Super Kings	bat	N	Rajasthan Royals	
386	980991	Chandigarh	2016-05-15	2016	46	Kings XI Punjab	Sunrisers Hyderabad	Punjab Cricket Association IS Bindra Stadium, ...	Kings XI Punjab	bat	N	Sunrisers Hyderabad	

```
# info
movies.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1629 entries, 0 to 1628
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x                1629 non-null  object
1   imdb_id                1629 non-null  object
2   poster_path            1526 non-null  object
3   wiki_link              1629 non-null  object
4   title_y                1629 non-null  object
5   original_title         1629 non-null  object
6   is_adult               1629 non-null  int64
7   year_of_release        1629 non-null  int64
8   runtime                1629 non-null  object
9   genres                 1629 non-null  object
10  imdb_rating            1629 non-null  float64
11  imdb_votes             1629 non-null  int64
12  story                  1609 non-null  object
13  summary                 1629 non-null  object
14  tagline                 557 non-null   object
15  actors                 1624 non-null  object
```

```

16 wins_nominations  707 non-null  object
17 release_date      1522 non-null object
dtypes: float64(1), int64(3), object(14)
memory usage: 229.2+ KB

```

```
ipl.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column              Non-Null Count  Dtype
---  -
0   ID                  950 non-null   int64
1   City                899 non-null   object
2   Date                950 non-null   object
3   Season              950 non-null   object
4   MatchNumber         950 non-null   object
5   Team1               950 non-null   object
6   Team2               950 non-null   object
7   Venue               950 non-null   object
8   TossWinner          950 non-null   object
9   TossDecision        950 non-null   object
10  SuperOver           946 non-null   object
11  WinningTeam         946 non-null   object
12  WonBy               950 non-null   object
13  Margin              932 non-null   float64
14  method              19 non-null    object
15  Player_of_Match     946 non-null   object
16  Team1Players        950 non-null   object
17  Team2Players        950 non-null   object
18  Umpire1             950 non-null   object
19  Umpire2             950 non-null   object
dtypes: float64(1), int64(1), object(18)
memory usage: 148.6+ KB

```

```

# describe
movies.describe()

```

```

is_adult  year_of_release  imdb_rating  imdb_votes
count    1629.0          1629.000000  1629.000000  1629.000000
mean      0.0            2010.263966   5.557459   5384.263352
std       0.0            5.381542    1.567609  14552.103231
min       0.0            2001.000000   0.000000   0.000000
25%       0.0            2005.000000   4.400000   233.000000
50%       0.0            2011.000000   5.600000  1000.000000
75%       0.0            2015.000000   6.800000  4287.000000
max       0.0            2019.000000   9.400000 310481.000000

```

```
ipl.describe()
```

```

ID      Margin
count  9.500000e+02  932.000000
mean   8.304852e+05   17.056867
std    3.375678e+05   21.633109
min    3.359820e+05    1.000000
25%    5.012612e+05    6.000000
50%    8.297380e+05    8.000000
75%    1.175372e+06   19.000000
max    1.312200e+06  146.000000

```

```

# isnull
movies.isnull().sum()

```

```

title_x      0
imdb_id      0
poster_path  103

```

```
wiki_link      0
title_y       0
original_title 0
is_adult       0
year_of_release 0
runtime       0
genres        0
imdb_rating    0
imdb_votes     0
story         20
summary       0
tagline       1072
actors        5
wins_nominations 922
release_date   107
dtype: int64
```

```
# duplicated
movies.duplicated().sum()
```

```
0
```

```
students.duplicated().sum()
```

```
1
```

```
# rename
students
```

```

  iq  percent  lpa
0  100      80   10
1   90      70    7
2  120     100   14
3   80      50    2
4    0       0    0
5    0       0    0
```

```
students.rename(columns={'marks': 'percent', 'package': 'lpa'}, inplace=True)
```

Math Methods

```
# sum -> axis argument
students.sum(axis=0)
```

```
iq      390
percent 300
lpa      33
dtype: int64
```

```
students.mean(axis=1)
```

```
0    63.333333
1    55.666667
2    78.000000
3    44.000000
4     0.000000
5     0.000000
dtype: float64
```

```
students.var()
```

```
iq      2710.0
percent 1760.0
lpa      33.5
dtype: float64
```

Start coding or [generate](#) with AI.

✓ Selecting cols from a DataFrame

```
# single cols
movies['title_x']
```

```
0          Uri: The Surgical Strike
1          Battalion 609
2    The Accidental Prime Minister (film)
3          Why Cheat India
4          Evening Shadows
...
1624        Tera Mera Saath Rahen
1625        Yeh Zindagi Ka Safar
1626          Sabse Bada Sukh
1627              Daaka
1628          Humsafar
Name: title_x, Length: 1629, dtype: object
```

```
ipl['Venue']
```

```
0          Narendra Modi Stadium, Ahmedabad
1          Narendra Modi Stadium, Ahmedabad
2          Eden Gardens, Kolkata
3          Eden Gardens, Kolkata
4          Wankhede Stadium, Mumbai
...
945          Eden Gardens
946          Wankhede Stadium
947          Feroz Shah Kotla
948    Punjab Cricket Association Stadium, Mohali
949          M Chinnaswamy Stadium
Name: Venue, Length: 950, dtype: object
```

```
# multiple cols
movies[['year_of_release', 'actors', 'title_x']]
```

```

year_of_release      actors      title_x
0          2019    Vicky Kaushal|Paresh Rawal|Mohit Raina|Yami Ga...    Uri: The Surgical Strike
1          2019    Vicky Ahuja|Shoaib Ibrahim|Shrikant Kamat|Elen...    Battalion 609
2          2019    Anupam Kher|Akshaye Khanna|Aahana Kumra|Atul S...    The Accidental Prime Minister (film)
3          2019    Emraan Hashmi|Shreya Dhanwanthary|Snighdadeep ...    Why Cheat India
4          2018    Mona Ambegaonkar|Ananth Narayan Mahadevan|Deva...    Evening Shadows
...          ...          ...          ...
1624        2001    Ajay Devgn|Sonali Bendre|Namrata Shirodkar|Pre...    Tera Mera Saath Rahen
1625        2001    Ameesha Patel|Jimmy Sheirgill|Nafisa Ali|Gulsh...    Yeh Zindagi Ka Safar
1626        2018    Vijay Arora|Asrani|Rajni Bala|Kumud Damle|Utpa...    Sabse Bada Sukh
1627        2019          Gippy Grewal|Zareen Khan|          Daaka
1628        2011          Fawad Khan|          Humsafar
1629 rows × 3 columns
```

```
ipl[['Team1', 'Team2', 'WinningTeam']]
```




	Team1	Team2	WinningTeam
0	Rajasthan Royals	Gujarat Titans	Gujarat Titans
1	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals
2	Royal Challengers Bangalore	Lucknow Super Giants	Royal Challengers Bangalore
3	Rajasthan Royals	Gujarat Titans	Gujarat Titans
4	Sunrisers Hyderabad	Punjab Kings	Punjab Kings
...	...	...	...
945	Kolkata Knight Riders	Deccan Chargers	Kolkata Knight Riders
946	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore
947	Delhi Daredevils	Rajasthan Royals	Delhi Daredevils
948	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings
949	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders

950 rows × 3 columns

✓ Selecting rows from a DataFrame

- **iloc** - searches using index positions
- **loc** - searches using index labels

```
# single row
movies.iloc[5]
```



```
title_x          Soni (film)
imdb_id          tt6078866
poster_path      https://upload.wikimedia.org/wikipedia/en/thum...
wiki_link        https://en.wikipedia.org/wiki/Soni_(film).
title_y          Soni
original_title    Soni
is_adult          0
year_of_release  2018
runtime          97
genres            Drama
imdb_rating       7.2
imdb_votes        1595
story            Soni a young policewoman in Delhi and her su...
summary          While fighting crimes against women in Delhi ...
tagline           NaN
actors            Geetika Vidya Ohlyan|Saloni Batra|Vikas Shukla...
wins_nominations  3 wins & 5 nominations
release_date      18 January 2019 (USA)
Name: 5, dtype: object
```

```
# multiple row
movies.iloc[:5]
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title	is_a
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thumb/Uri:_The_Surgical_Strike/Uri:_The_Surgical_Strike.jpg	https://en.wikipedia.org/wiki/Uri:_The_Surgical_Strike	Uri: The Surgical Strike	Uri: The Surgical Strike	
1	Battalion 609	tt9472208	NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609	
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thumb/The_Accidental_Prime_Minister_(film)/The_Accidental_Prime_Minister_(film).jpg	https://en.wikipedia.org/wiki/The_Accidental_Prime_Minister_(film)	The Accidental Prime Minister	The Accidental Prime Minister	
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thumb/Why_Cheat_India/Why_Cheat_India.jpg	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India	
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows	

```
# fancy indexing
movies.iloc[[0,4,5]]
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title	is_adult
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thumb/Uri:_The_Surgical_Strike/Uri:_The_Surgical_Strike.jpg	https://en.wikipedia.org/wiki/Uri:_The_Surgical_Strike	Uri: The Surgical Strike	Uri: The Surgical Strike	
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows	
5	Soni (film)	tt6078866	https://upload.wikimedia.org/wikipedia/en/thumb/Soni_(film)/Soni_(film).jpg	https://en.wikipedia.org/wiki/Soni_(film)	Soni	Soni	

```
# loc
students
```



	iq	marks	package
name			
nitish	100	80	10
ankit	90	70	7
rupesh	120	100	14
rishabh	80	50	2
amit	0	0	0
ankita	0	0	0

```
students.loc['nitish']
```



iq	100
marks	80
package	10
Name: nitish, dtype: int64	

```
students.loc['nitish':'rishabh':2]
```



	iq	marks	package
name			
nitish	100	80	10
rupesh	120	100	14

```
students.loc[['nitish','ankita','rupesh']]
```



	iq	marks	package
name			
nitish	100	80	10
ankita	0	0	0
rupesh	120	100	14

```
students.iloc[[0,3,4]]
```



	iq	marks	package
name			
nitish	100	80	10
rishabh	80	50	2
amit	0	0	0

▼ Selecting both rows and cols

```
movies.iloc[0:3,0:3]
```



	title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...
1	Battalion 609	tt9472208	NaN
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...

```
movies.loc[0:2, 'title_x': 'poster_path']
```



	title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...
1	Battalion 609	tt9472208	NaN
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...

Filtering a DataFrame

```
ipl.head(2)
```



	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	SuperOver	WinningTeam	W
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat	N	Gujarat Titans	W
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field	N	Rajasthan Royals	W

```
# find all the final winners
mask = ipl['MatchNumber'] == 'Final'
new_df = ipl[mask]
new_df[['Season', 'WinningTeam']]
```

```
ipl[ipl['MatchNumber'] == 'Final'][['Season', 'WinningTeam']]
```



	Season	WinningTeam
0	2022	Gujarat Titans
74	2021	Chennai Super Kings
134	2020/21	Mumbai Indians
194	2019	Mumbai Indians
254	2018	Chennai Super Kings
314	2017	Mumbai Indians
373	2016	Sunrisers Hyderabad
433	2015	Mumbai Indians
492	2014	Kolkata Knight Riders
552	2013	Mumbai Indians
628	2012	Kolkata Knight Riders
702	2011	Chennai Super Kings
775	2009/10	Chennai Super Kings
835	2009	Deccan Chargers
892	2007/08	Rajasthan Royals

```
# how many super over finishes have occurred
ipl[ipl['SuperOver'] == 'Y'].shape[0]
```



```
14
```

```
# how many matches has csk won in kolkata
ipl[(ipl['City'] == 'Kolkata') & (ipl['WinningTeam'] == 'Chennai Super Kings')].shape[0]
```



```
5
```

```
# toss winner is match winner in percentage
(ip1[ip1['TossWinner'] == ip1['WinningTeam']].shape[0]/ip1.shape[0])*100
```

51.473684210526315

```
# movies with rating higher than 8 and votes>10000
movies[(movies['imdb_rating'] > 8.5) & (movies['imdb_votes'] > 10000)].shape[0]
```

0

```
# Action movies with rating higher than 7.5
# mask1 = movies['genres'].str.split('|').apply(lambda x:'Action' in x)
mask1 = movies['genres'].str.contains('Action')
mask2 = movies['imdb_rating'] > 7.5
```

```
movies[mask1 & mask2]
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The S
41	Family of Thakurganj	tt8897986	https://upload.wikimedia.org/wikipedia/en/9/99...	https://en.wikipedia.org/wiki/Family_of_Thakur...	Family of Thakurganj	F. Tha
84	Mukkabaaz	tt7180544	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Mukkabaaz	The Brawler	Muk
106	Raazi	tt7098658	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Raazi	Raazi	
110	Parmanu: The Story of Pokhran	tt6826438	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Parmanu:_The_Sto...	Parmanu: The Story of Pokhran	Parma Story of F
112	Bhavesh Joshi Superhero	tt6129302	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Bhavesh_Joshi_Su...	Bhavesh Joshi Superhero	Bhaves Su
169	The Ghazi Attack	tt6299040	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Ghazi_Attack...	The Ghazi Attack	The Ghaz
219	Raag Desh (film)	tt6080746	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Raagdes	Raag Desh	Rae
258	Irudhi Suttru	tt5310090	https://upload.wikimedia.org/wikipedia/en/ff/fe...	https://en.wikipedia.org/wiki/Saala_Khadoos	Saala Khadoos	Saala K
280	Laal Rang	tt5600714	NaN	https://en.wikipedia.org/wiki/Laal_Rang	Laal Rang	La:

```
# write a function that can return the track record of 2 teams against each other
```

✓ Adding new cols

```
# completely new
movies['Country'] = 'India'
movies.head()
```

		title_x	imdb_id	poster_path	wiki_link	title_y	original_title	is_a
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...	Uri: The Surgical Strike	Uri: The Surgical Strike		
1	Battalion 609	tt9472208		NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609	
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister		
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India		
4	Evening Shadows	tt6028796		NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows	

```
# from existing ones
movies.dropna(inplace=True)
```

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```

title_x    imdb_id    poster_path    wiki_link    title_y    original_title    is_adul
11    Gully Boy    tt2395469    https://upload.wikimedia.org/wikipedia/en/thum...    https://en.wikipedia.org/wiki/Gully_Boy    Gully Boy    Gully Boy    1
movies.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 298 entries, 11 to 1623
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x                298 non-null   object
1   imdb_id                298 non-null   object
2   poster_path            298 non-null   object
3   wiki_link              298 non-null   object
4   title_y                298 non-null   object
5   original_title         298 non-null   object
6   is_adult               298 non-null   int64
7   year_of_release        298 non-null   int64
8   runtime                298 non-null   object
9   genres                 298 non-null   object
10  imdb_rating            298 non-null   float64
11  imdb_votes             298 non-null   int64
12  story                  298 non-null   object
13  summary                298 non-null   object
14  tagline                 298 non-null   object
15  actors                 298 non-null   object
16  wins_nominations        298 non-null   object
17  release_date           298 non-null   object
18  Country                 298 non-null   object
dtypes: float64(1), int64(3), object(15)
memory usage: 46.6+ KB

```

▼ Important DataFrame Functions

```

# astype
ipl.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    950 non-null   int64
1   City                  899 non-null   object
2   Date                  950 non-null   object
3   Season                950 non-null   object
4   MatchNumber           950 non-null   object
5   Team1                 950 non-null   object
6   Team2                 950 non-null   object
7   Venue                 950 non-null   object
8   TossWinner            950 non-null   object
9   TossDecision          950 non-null   object
10  SuperOver             946 non-null   object
11  WinningTeam           946 non-null   object
12  WonBy                 950 non-null   object
13  Margin                932 non-null   float64
14  method                19 non-null    object
15  Player_of_Match       946 non-null   object
16  Team1Players           950 non-null   object
17  Team2Players           950 non-null   object
18  Umpire1               950 non-null   object
19  Umpire2               950 non-null   object
dtypes: float64(1), int64(1), object(18)
memory usage: 148.6+ KB

```

```
ipl['ID'] = ipl['ID'].astype('int32')
```

```
ipl.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):

```